# Probability & Statistics

ClipSAT Printable Practice Suite (clipsat.org) • Topic 11.01

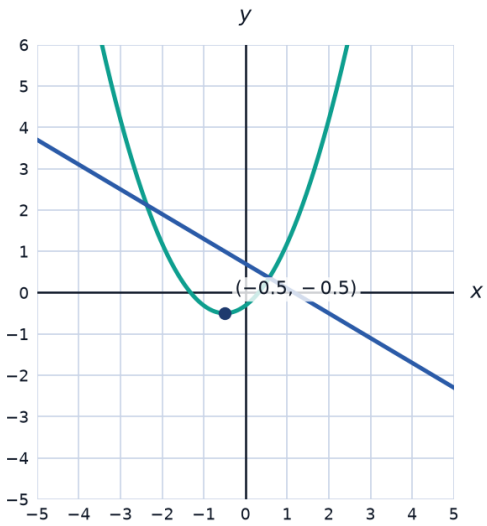
## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

What is the value of $16^{\frac{3}{4}}$?

**2** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Which expression is equivalent to $(16x^8y^4)^{\frac{3}{4}}$ for all positive values of $x$ and $y$?




**3** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

What are the solutions to the quadratic equation $x^2 - 6x + 13 = 0$?

**4** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

What are the center and radius of the circle given by the equation $(x - 2)^2 + (y + 6)^2 = 36$?

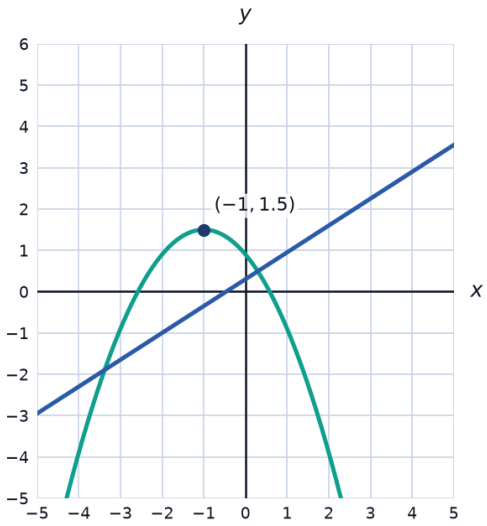
**5** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A sequence is defined by the recursive formula $a_n = 2a_{n-1} + 1$, with the first term $a_1 = 3$. What is the value of the third term, $a_3$?

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**6** [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the quadratic equation $x^2 - 8x + 15 = 0$. What are the roots of this equation?



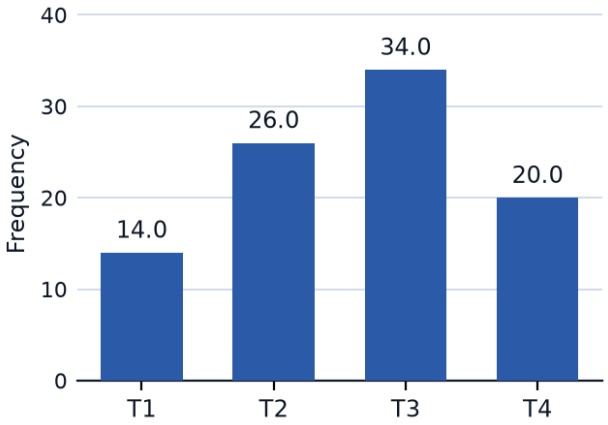

**7** [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

What is the center of the circle represented by the equation $(x - 3)^2 + (y + 5)^2 = 16$?

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

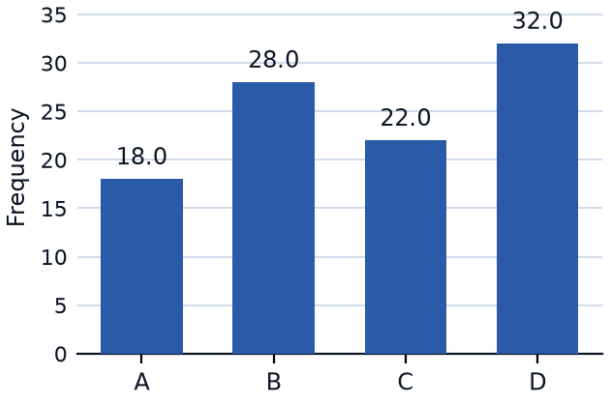
For the model $f(x) = 3x + 4$ in describing shapes of data distributions, evaluate $f(5)$.




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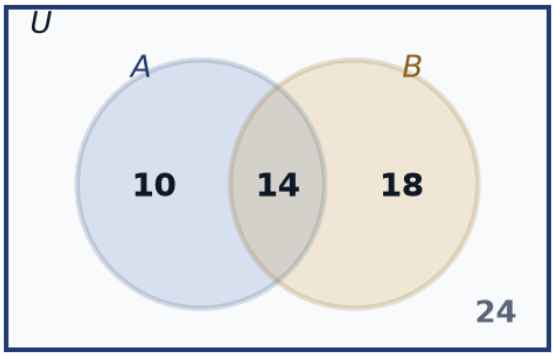
**9** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 4x + 5$ in the normal distribution, evaluate $f(2)$.




**10** [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A data set recorded for the normal distribution consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean? Show your complete algebraic work.




**11** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A committee of 4 members is to be selected at random from a club of 7 seniors and 5 juniors. (a) What is the total number of possible committees? (b) Find the probability that the committee contains exactly 2 seniors and 2 juniors.

**12** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

[Probability & Statistics · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input $t$ when $g(t) = 50$.

**13** [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In a class of 200 students taking a test with mean 65 and SD 12 (normally distributed), how many students would you expect to score above 80?

**14** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Solve the rational equation for $x$: $\frac{x}{x-2} + \frac{1}{x+3} = \frac{5}{x^2+x-6}$. Check for extraneous solutions.

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## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

**15** **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 6.01 Radicals & Rational Exponents**

Solve the linear system using matrix inversion or Cramer's Rule: $3x - 2y = 8$ $5x + 4y = 6$

**16** **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)** • **Cross-Lesson: 7.01 Sequences & Series**

Find the inverse function $f^{-1}(x)$ and state its domain and range for $f(x) = \frac{3x-1}{2x+5}$.




**17** **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)** • **Cross-Lesson: 8.01 Conic Sections**

If $\sin\theta = -\frac{4}{5}$ and $\theta$ lies in Quadrant III, find the exact values of $\cos\theta$, $\tan\theta$, and $\sin(2\theta)$.

**18** **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)** • **Cross-Lesson: 9.01 Trigonometric Functions**

Solve $2x^2 - 6x + 5 = 0$ by completing the square, expressing all solutions in exact $a \pm bi$ form.

**19** **[Challenging]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)** • **Cross-Lesson: 1.01 Functions & Their Graphs**

Find all real and complex roots of the polynomial $P(x) = 2x^3 + 3x^2 - 8x + 3$.

**20** **[Challenging]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)** • **Cross-Lesson: 10.01 Matrices & Systems**

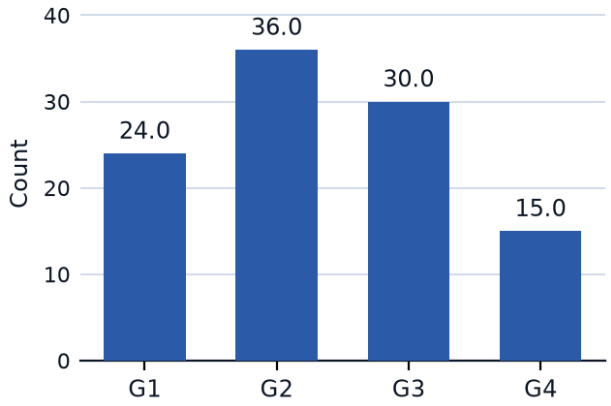
Solve the radical equation $\sqrt{2x+7} - x = 2$ for all real solutions and confirm extraneous roots.

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## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 6x + 9$ in random sampling, evaluate $f(2)$.




**a.** 21

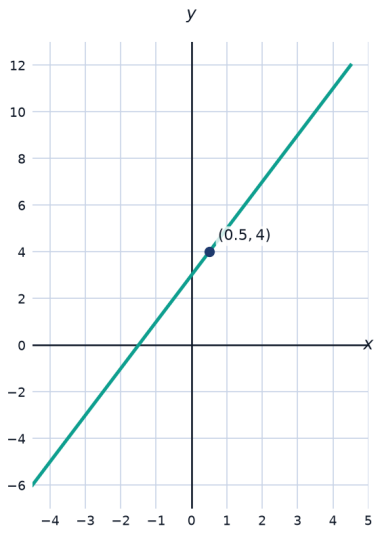
**b.** 27

**c.** 12

**d.** 25

**22** [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 2x + 3$ in estimating proportions from samples, evaluate $f(2)$.




**a.** 7

**b.** 9

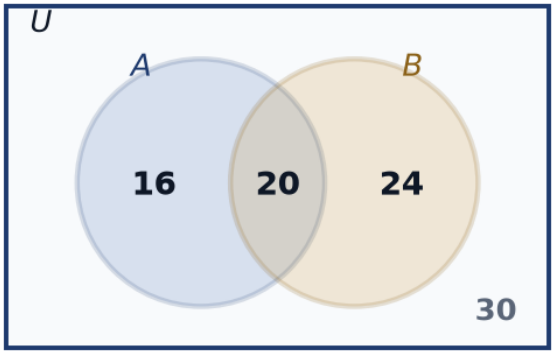
**c.** 4

**d.** 11

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**23** [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for random sampling consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?




**a.** 20.5

**b.** 21.5

**c.** 22.5

**d.** 24.5

**24** [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for estimating proportions from samples consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

**a.** 21.5

**b.** 22.5

**c.** 23.5

**d.** 25.5

**25** [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for describing shapes of data distributions consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

**a.** 22.5

**b.** 23.5

**c.** 24.5

**d.** 26.5

**26** [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for interpreting and comparing z-scores consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

**a.** 15.5

**b.** 16.5

**c.** 17.5

**d.** 19.5

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**27** [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 5x + 6$ in interpreting and comparing z-scores, evaluate $f(3)$.

**a.** 21

**b.** 26

**c.** 15

**d.** 25

**28** [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for normal distribution probabilities consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

**a.** 16.5

**b.** 17.5

**c.** 18.5

**d.** 20.5

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 6x + 7$ in normal distribution probabilities, evaluate $f(3)$.

**a.** 25

**b.** 31

**c.** 18

**d.** 29

**30** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A fair six-sided die is rolled twice. Let $X$ be the random variable representing the sum of the outcomes of the two rolls. The probability distribution of $X$ is given in the table below, where $p$ is an unknown probability. $Sum\ (x), 2, 3, 4, 5, 6, 7; P(X = x), 136, p, 336, 436, 536, 636;$ What is the value of $p$?

**a.** $\frac{2}{36}$

**b.** $\frac{5}{36}$

**c.** $\frac{4}{36}$

**d.** $\frac{3}{36}$

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**31** **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

The weights of apples harvested in an orchard are approximately normally distributed with mean $\mu$ and standard deviation $\sigma$. According to the empirical rule (68-95-99.7% rule), approximately 84% of the apples have a weight greater than 166 grams, and approximately 2.5% of the apples have a weight greater than 214 grams. What is the mean weight $\mu$, in grams, of the apples?

**a.** 178 grams

**b.** 182 grams

**c.** 190 grams

**d.** 198 grams

**32** **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

In a championship playoff, two teams, Team A and Team B, play a best-of-five series, where the first team to win three games wins the series. The games are independent, cannot end in a tie, and in each individual game, the probability that Team A wins is $p$, where $0 < p < 1$ and $p \neq \frac{1}{2}$. Suppose the probability that the series is decided in exactly 3 games is equal to the probability that the series requires a decisive 5th game. What is the value of the product $p(1-p)$?

**a.** $\frac{\sqrt{33}-3}{12}$

**b.** $\frac{\sqrt{33}+3}{12}$

**c.** $\frac{\sqrt{33}-3}{6}$

**d.** $\frac{6-\sqrt{33}}{12}$

**33** **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

A set of test scores is normally distributed with a mean of $\mu = 500$ and a standard deviation of $\sigma = 100$. According to the empirical rule (the $68--95--99.7\%$ rule), approximately what percentage of scores fall between $400$ and $700$?

**a.** $81.5\%$

**b.** $68.0\%$

**c.** $84.0\%$

**d.** $95.0\%$

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**34** [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A factory produces two types of components, A and B, using two machines, M1 and M2. The time (in minutes) required to produce one unit of each component on each machine is given in the table below: | Component | Machine M1 | Machine M2 | |---|---|---| | A | 2 | 3 | | B | 4 | 1 | The factory has 480 minutes of M1 time and 240 minutes of M2 time available per day. Let $x$ be the number of units of component A produced per day and $y$ be the number of units of component B produced per day. Suppose the profit from selling one unit of component A is $5$ and the profit from selling one unit of component B is $8$. The factory wants to maximize its total daily profit. Formulate this problem as a linear programming problem, defining the objective function and the constraints.

**a.** Maximize $P = 5x + 8y$ subject to $2x + 4y \leq 480$, $3x + y \leq 240$, $x \geq 0$, $y \geq 0$.

**b.** Maximize $P = 8x + 5y$ subject to $2x + 3y \leq 480$, $4x + y \leq 240$, $x \geq 0$, $y \geq 0$.

**c.** Maximize $P = 5x + 8y$ subject to $4x + 2y \leq 480$, $y + 3x \leq 240$, $x \geq 0$, $y \geq 0$.

**d.** Maximize $P = 5x + 8y$ subject to $2x + 4y \leq 240$, $3x + y \leq 480$, $x \geq 0$, $y \geq 0$.

**35** [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A bag contains 5 red marbles and 3 blue marbles. If one marble is drawn at random, what is the probability that the marble is red?

**a.** $\frac{3}{8}$

**b.** $\frac{5}{8}$

**c.** $\frac{1}{2}$

**d.** $\frac{5}{3}$

**36** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Prove the weak law of large numbers: $\bar{X}_n \to \mu$ as $n \to \infty$.

**a.** By Chebyshev's inequality: $P(|\bar{X}_n - \mu| > \varepsilon) \leq \frac{\text{Var}(\bar{X}_n)}{\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2} \to 0$. ✓

**b.** Not provable

**c.** Use the CLT only

**d.** Only for normal distributions

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**37** [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The moment generating function uniquely determines the distribution. Why?

**a.** Only if $E(X)$ is finite

**b.** Not always true

**c.** Only for discrete distributions

**d.** Because the MGF encodes all moments and there is a one-to-one correspondence between MGFs and distributions (under regularity conditions)

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## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 9.01 Trigonometric Functions

State and prove the Bayes' theorem for continuous distributions.

**a.** Not valid for continuous distributions

**b.** Same as discrete case

**c.** $f(\theta|x) = \frac{f(x|\theta)\pi(\theta)}{\int f(x|\theta)\pi(\theta)d\theta}$ — from the definition of conditional density

**d.** Use the CLT

**39** [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.01 Functions & Their Graphs

Show that the binomial distribution approaches Poisson as $n \to \infty$, $p \to 0$, $np = \lambda$ fixed.

**a.** Requires $n > 100$

**b.** Not true

**c.** $\binom{n}{k}p^k(1-p)^{n-k} \to \frac{e^{-\lambda}\lambda^k}{k!}$ — use $(1-\lambda/n)^n \to e^{-\lambda}$

**d.** Only for $k = 0$

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**40** [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 10.01 Matrices & Systems

State the definition of a sufficient statistic.

**a.** $T(X)$ is sufficient for $\theta$ if the conditional distribution of $X$ given $T(X)$ does not depend on $\theta$

**b.** It must be unbiased

**c.** $T(X) = \theta$ always

**d.** Only the sample mean is sufficient

**41** [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Quadratics & Complex Numbers

State and apply the Central Limit Theorem.

**a.** Only for discrete distributions

**b.** For large $n$, $\bar{X} \sim N(\mu, \sigma^2/n)$ approximately, regardless of the population distribution

**c.** Only for $n > 1000$

**d.** Only for normal populations

**42** [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.01 Polynomial Functions

In a hypothesis test, the $p$-value is:

**a.** The probability that $H_0$ is true

**b.** The probability of observing a test statistic as extreme as observed, assuming $H_0$

**c.** The type II error rate

**d.** The significance level

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**43** [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 4.01 Rational Expressions

State the Fisher information: $I(\theta) =$

**a.** $E\left[\frac{\partial^2}{\partial\theta^2}\ln f\right]$

**b.** $E\left[\left(\frac{\partial}{\partial\theta}\ln f(X;\theta)\right)^2\right]$

**c.** $\frac{1}{\mathrm{Var}(\hat{\theta})}$

**d.** $\mathrm{Var}(\hat{\theta})$

**44** [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

Show the mgf of $\bar{X}$ for i.i.d. $X_i$ with mgf $M(t)$ is $[M(t/n)]^n$.

**a.** $M_{\bar{X}}(t) = E[e^{t\bar{X}}] = E\left[\prod e^{tX_i/n}\right] = \prod M(t/n) = [M(t/n)]^n$

**b.** Only for normal $X$

**c.** Only for $n = 2$

**d.** Not true

**45** [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 Radicals & Rational Exponents

For two random variables $X, Y$, prove $\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\mathrm{Cov}(X,Y)$.

**a.** Only if independent

**b.** Only for discrete variables

**c.** $E[(X+Y-\mu_X-\mu_Y)^2] = E[(X-\mu_X)^2] + 2E[(X-\mu_X)(Y-\mu_Y)] + E[(Y-\mu_Y)^2]$. ✓

**d.** Not true

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